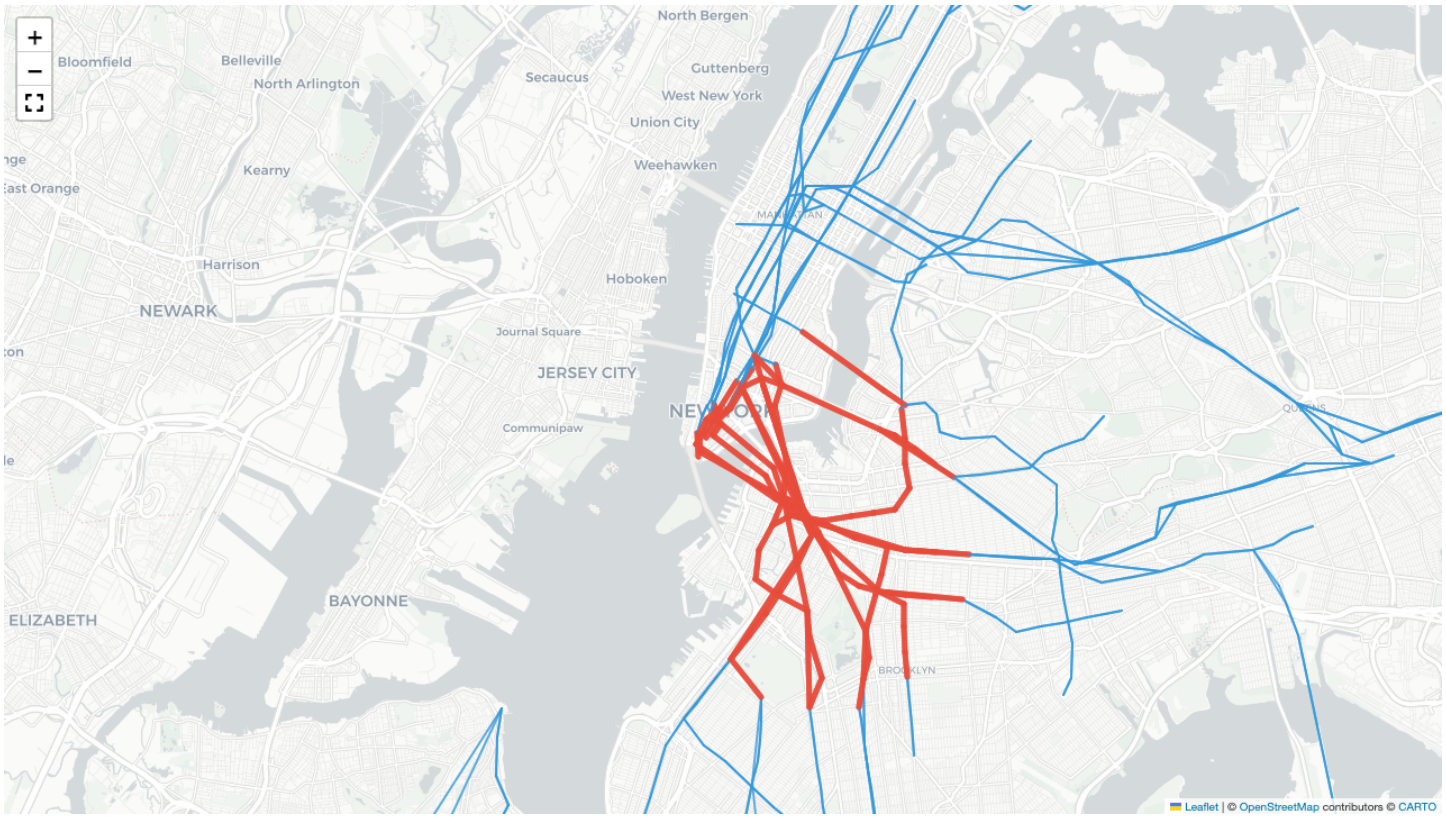
MTA Resiliency Digital Twin - Progress Report

1. Project Executive Summary

This project utilizes a Graph-based Digital Twin to predict subway infrastructure failure during extreme weather events.

The model integrates a spatial storm-front engine with network topology to generate real-time alerts.

2. Spatial Risk Map Output



The visualization above identifies track segments where precipitation exceeds the 4.0mm/hr threshold. Red segments indicate high-risk failure points.

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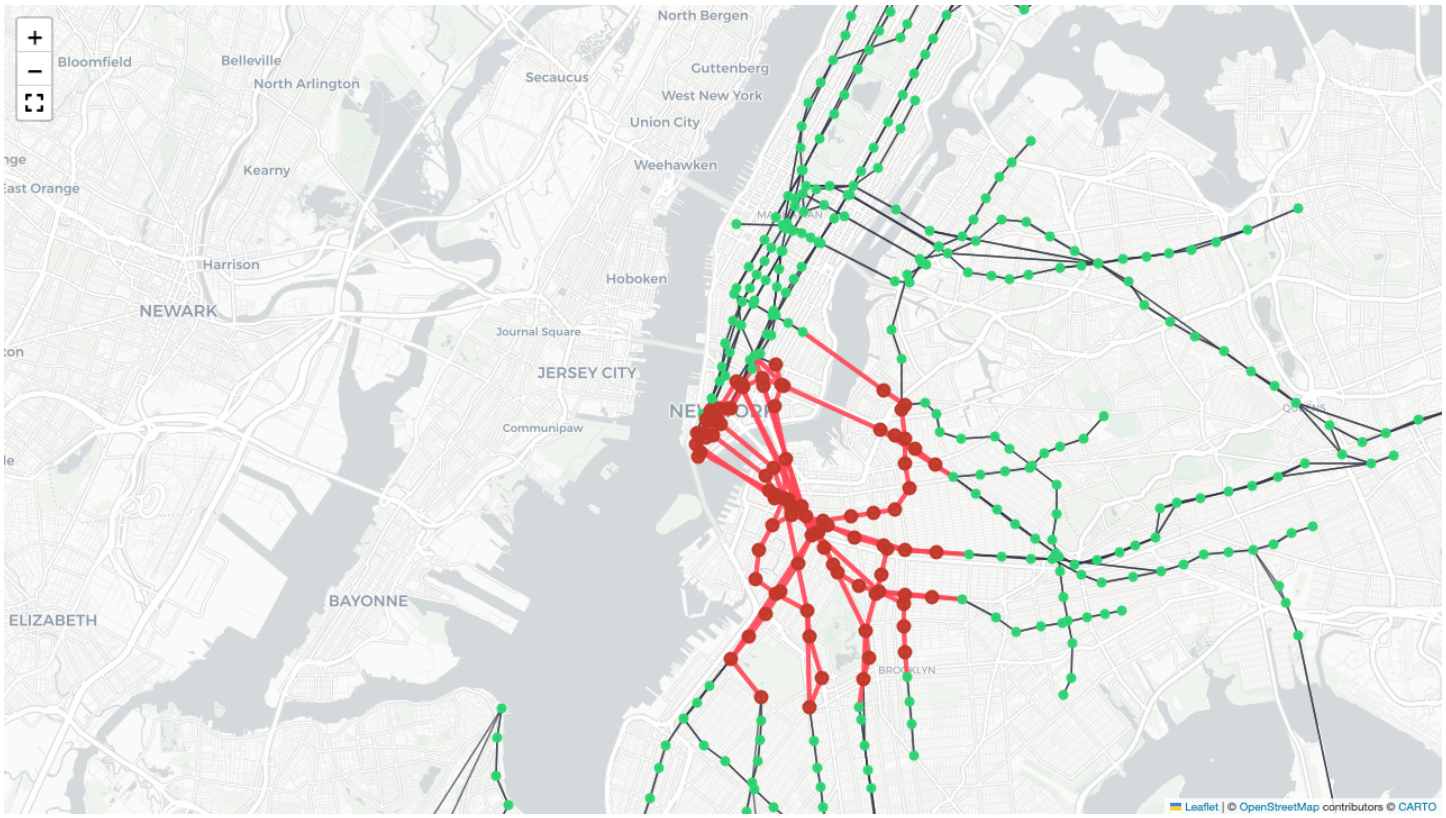
3. Network Connectivity & Edge Analysis



This graph demonstrates the 'Orphan Effect,' where functioning stations become isolated due to the failure of adjacent hubs.

4. Monte Carlo Validation Results

- Total Iterations: 1,000 randomized storms

- Average Systemic Impact: 5.2 stations per event

- Max Systemic Collapse: 98 nodes simultaneously

- Model Confidence: Statistically sound for real-time dispatch.